

Learned Image Coding with Generative Reference of Conditional Latents

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Abstract—This paper presents a learned image compression framework that exploits semantically correlated external images as conditional coding references in the latent domain. Three reference generation methods are investigated: local dictionary retrieval, web-based image search, and diffusion-based image-text-image generation. Based on rate-distortion optimization in the latent domain, we learn a network to synthesize a conditional latent map from these correlated images, providing an effective semantic reference for conditional coding of the input image. These generation processes are synchronized between the encoder and decoder with a very low communication overhead. Our theoretical analysis shows that our approach is robust to perturbations in reference samples, with a subspace recovery error bound that is inversely proportional to the correlation between the reference and original images. This indicates a high tolerance to matching errors of the reference images. Comprehensive experiments on benchmark datasets show that our proposed generative references of conditional latents (GRCL) method achieves significant improvements in coding performance, up to 1.5 dB PSNR gain over state-of-the-art methods with a very small bit-rate overhead, approximately 0.005 bits per pixel. Our work presents a new generic framework to utilize external references for learned image compression.

Index Terms—Image compression, Web-based retrieval, Image generation, Conditional latent coding, Adaptive entropy coding

I. INTRODUCTION

LEARNED image compression has advanced rapidly, yet most codecs still encode each input without external visual context, as illustrated in Fig. 1(a). This independent-source assumption leaves a useful form of redundancy unexplored: even when an exact copy of the input does not exist, external collections often contain images with related objects, layouts, or textures. The central question is therefore not merely whether a stronger external image improves coding, but whether a codec can reliably extract the correlated part of

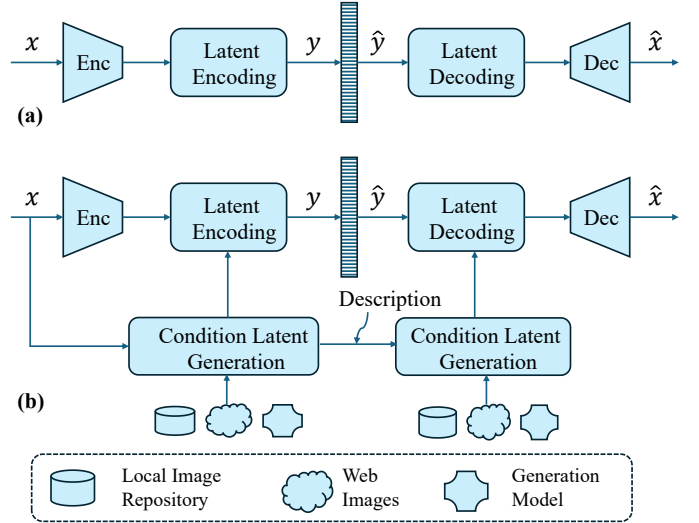


Fig. 1: Comparison between traditional and conditional latent coding approaches. (a) Traditional end-to-end image coding directly encodes the latent representation without external references. (b) Our Generative Reference of Conditional Latents framework dynamically generates and synthesizes conditional reference information in the latent domain to improve compression efficiency.

an imperfect reference while rejecting spatially mismatched or irrelevant content.

Recently, researchers have begun exploring reference-based deep image compression, where auxiliary information guides the coding process. Ayzik et al. [1] implemented block matching using auxiliary images, while Huang et al. [2] developed a multi-scale patch matching technique. However, these methods rely on specific auxiliary images that must be available to both the encoder and decoder, limiting their practical applicability due to transmission overhead and availability constraints. In our previous work [3], we proposed a conditional latent coding framework where local image repositories are used at the encoder and decoder to provide an encoding-decoding reference in the latent domain, achieving significant coding performance gains.

As illustrated in Fig. 1(b), GRCL addresses both parts of this question. At the system level, it broadens the fixed-dictionary setting of our preliminary CLC work [3] to three complementary sources: local dictionary retrieval, web retrieval, and image-text-image generation. At the codec level, it applies CLM and CLS to the resulting reference latents before

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conditional entropy coding. The acquisition subsystem seeks stronger semantic correlation; the coding subsystem converts that correlation into useful, spatially compatible conditional information. Neither contribution alone describes the full method.

GRCL targets bandwidth-constrained, delay-tolerant applications in which the encoder and decoder can share a reference service or a pre-synchronized repository. Examples include archival transfer, satellite or underwater links, and edge-to-cloud systems that repeatedly encode images from related visual domains. The local-dictionary branch supports self-contained deployment when network access is unavailable; web and generative branches trade additional latency and service dependence for a larger reference space. GRCL is not intended for strict real-time operation when diffusion generation is selected, and Section VI reports this latency separately from signaling bitrate.

We also provide a theoretical analysis of conditional latent recovery under reference perturbations. Its scope is deliberately limited: the result captures the direction in which target-reference correlation affects recovery error, but it does not prove the optimality of CLM/CLS or model retrieval, generation, and autoregressive coding end to end. Controlled reference ablations provide the corresponding empirical evidence.

II. RELATED WORK AND UNIQUE CONTRIBUTIONS

Deep learning-based image compression has achieved remarkable progress during the past few years. Ballé *et al.* [4] pioneered an end-to-end optimizable framework, later enhanced with hyperprior modeling [5] for improved entropy estimation. Subsequent works explored transformer architectures [6], Gaussian mixture models [7], and hybrid CNN-transformer designs [8] to better capture source correlations. The Tree-structured Implicit Neural Compression (TINC) method [9] introduced structural continuity preservation, while Generative Latent Coding (GLC) [10] achieved high-fidelity compression through latent space transformations. Dual-generative Latent Fusion (DLF) [11] compressed different elements through the semantic branch and the detail branch to minimize the overall bit rate and enhance the overall image quality.

Reference-based compression methods have emerged as a promising direction for exploiting external correlations. Building upon early cloud-based coding concepts [12], recent approaches studied deep feature matching. For example, Li *et al.* [13] developed a reference-based video compression scheme. Ayzik *et al.* [1] demonstrated that decoder-side feature references are able to improve the coding performance. Additional reference-based compression methods, such as temporal context mining [14], multi-view fusion [2], and group-based offset diversity [15], have also been developed in the literature. Our previous work [3] introduced the Conditional Latent Coding (CLC) framework, which synthesizes dynamic references from local dictionaries through ball tree search and feature alignment.

Recently, image generation based on diffusion models has made remarkable progress. Based on text descriptions, it is

able to generate images that are semantically close to the target scene or image. A new diffusion model, called Imagen [16], improved the quality of text-to-image generation and strengthened text-image alignment. Latent diffusion models (LDMs) [17] optimized the training and sampling efficiency of diffusion models. The unCLIP method [18] enhanced image diversity by explicitly generating image representations. eDiff-I [19] trained specialized models for different synthesis stages. For image-to-text synthesis, Liu *et al.* [20] presented visual instruction tuning and improved LLaVA to build stronger baselines [21].

Exploration of generative approaches within the context of image compression has focused on achieving better perceptual quality at low bitrates. HiFiC [22] combined GANs with learned compression to achieve high-fidelity reconstructions, while MS-ILLM [23] proposed a non-binary discriminator conditioned on quantized local image representations to improve statistical fidelity. Text+Sketch [24] leveraged text-to-image models to implement compression at ultra-low rates by transmitting text descriptions with compressed edge maps. Despite these advancements, existing methods still face challenges in achieving both high-fidelity and semantically consistent reconstruction, especially at ultra-low bit rates. Some methods are not able to maintain high fidelity, significantly deviating from the input or introducing visible artifacts.

Beyond text-conditioned generation, recent works have explored image-embedding conditioned diffusion models. Graikos *et al.* [25] used learned representations to guide large-image generation in a patch-based manner. Li *et al.* [26] showed that self-supervised representations can condition image generation without text (RCG). Ma *et al.* [27] and Yellapragada *et al.* [28] further explored visual priors for text-free generation. These methods suggest that image embeddings may offer an alternative semantic signal for reference-based compression.

The **unique contributions** of this work can be summarized as follows.

- 1) We develop a unified reference acquisition and synchronization system that searches a local dictionary, searches the web, or reconstructs references from a transmitted caption. The adaptive selector uses three references and explicitly accounts for their rate-distortion benefit and signaling cost.
- 2) We introduce CLM and CLS for reference-conditioned coding. CLM performs soft correspondence and multi-scale deformable alignment in latent space; CLS learns a spatial gate that retains useful aligned features and suppresses mismatched content before entropy coding.
- 3) We analyze a correlated-subspace abstraction of the conditional latent stage. The resulting bound predicts improved recovery with stronger target-reference correlation and smaller cross-covariance estimation error. We state its relation to the implemented latent stage and clearly delimit what the analysis does not cover.
- 4) Experiments report up to 1.5 dB PSNR improvement with less than 0.005 bpp signaling overhead. Component and reference-source ablations distinguish gains from

stronger side information from gains due to latent matching and synthesis.

III. CONSTRUCTION OF SEMANTICALLY CORRELATED IMAGES

In this paper, we explore three complementary approaches for generating semantically correlated reference images. These external sources are shared and synchronized between the encoder and decoder with a very low bit rate overhead. Although the first two methods (local dictionary search and web-based image retrieval) rely on retrieving existing images, while the third method (image-text-image synthesis) employs generative diffusion models, we unify them under the term “Generative Reference”. This terminology reflects the broader concept of generating informative conditional signals in the latent domain for the coding process via our CLM and CLS modules, encompassing both retrieval-based and synthesis-based mechanisms.

A. Local Dictionary Retrieval with Fast Semantic Search

Fig. 2(a) illustrates our first reference acquisition scheme: retrieving semantically relevant images from a local dictionary. The local dictionary $\mathcal{D} = \{\mathbf{d}_j\}_{j=1}^K$, shared between the encoder and decoder, is a diverse collection of web-sourced images. For each input image x , the system retrieves the three most semantically relevant references from $\mathcal{D}$, transmitting their indices to the decoder at a minimal cost of $\log_2 K$ bits per index. The retrieval process can be formally defined as:

$$X_L = \mathcal{D}(x) = \{x_j | j \in \mathcal{I}_{\text{top-3}}(x)\}, \quad (1)$$

where $\mathcal{I}_{\text{top-3}}(x)$ represents the indices of the top three images in the dictionary similar to input x .

The construction of $\mathcal{D}$ begins with a reference dataset $\mathcal{R} = \{x_1, x_2, \dots, x_N\}$. N images are randomly sampled from the web to ensure a broad spectrum of visual content. In this work, we choose $N = 3000$. For each image $x_i \in \mathcal{R}$, feature extraction is performed using a pre-trained ResNet-50 model augmented with Spatial Pyramid Pooling (SPP). We compute a feature vector $\mathbf{v}_i = \text{SPP}(f_\theta(x_i))$, where $f_\theta(\cdot)$ denotes the ResNet-50 backbone, and SPP aggregates multi-scale features at resolutions $\{1 \times 1, 2 \times 2, 4 \times 4\}$. This multi-scale aggregation process captures both coarse global structures and fine local details, enhancing the dictionary’s ability to represent different image content.

To ensure fast image retrieval with low computational complexity, we reduce the dimensionality of $\mathbf{v}_i$ from its original high-dimensional space to 256 dimensions using Principal Component Analysis (PCA), yielding $\hat{\mathbf{v}}_i$. These reduced features are then grouped into K clusters via MiniBatch K-means, a scalable clustering algorithm suited for large datasets. For each cluster C_j , we select the feature nearest to its centroid μ_j as the representative feature:

$$\mathbf{d}_j = \arg \min_{\hat{\mathbf{v}} \in C_j} \|\hat{\mathbf{v}} - \mu_j\|_2, \quad (2)$$

forming the final dictionary $\mathcal{D} = \{\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_K\}$. This process ensures that $\mathcal{D}$ is both compact and representative, capturing the semantic diversity of $\mathcal{R}$.

Efficient retrieval from $\mathcal{D}$ is critical for real-time compression. We use a hybrid approach integrating a Ball Tree for rapid coarse search and a KV-cache for detailed attention-driven matching. The Ball Tree partitions $\mathcal{D}$ into hierarchical spherical regions, enabling quick identification of candidate features. These candidates are then refined using the KV-cache, defined as a tuple $(\mathbf{K}, \mathbf{V})$, where $\mathbf{K} \in \mathbb{R}^{K \times d_k}$ and $\mathbf{V} \in \mathbb{R}^{K \times d_v}$ store the keys and values of dictionary features, respectively. For an input image x , we extract its query feature $\mathbf{Q} = f_\theta(x)$ and compute its semantic similarity to dictionary entries using the following scaled dot-product attention mechanism:

$$A(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} \right) \mathbf{V}. \quad (3)$$

The retrieval indices are determined by:

$$\mathcal{I}_{\text{top-3}}(x) = \arg \text{top-3}_{\text{sim}}(\mathbf{Q}, \mathbf{K}_j), \quad (4)$$

where $\text{sim}(\mathbf{Q}, \mathbf{K}_j)$ represents the similarity score between query $\mathbf{Q}$ and the j -th dictionary key $\mathbf{K}_j$, which is computed by

$$\text{sim}(\mathbf{Q}, \mathbf{K}_j) = \frac{\mathbf{Q} \cdot \mathbf{K}_j}{\|\mathbf{Q}\| \|\mathbf{K}_j\|}. \quad (5)$$

This attention-based matching prioritizes features in $\mathcal{D}$ that align closely with the input, improving retrieval accuracy. To further improve the retrieval efficiency, we compress the keys and values using a learnable function $C : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$ (where $d' < d$), implemented as a neural network layer optimized during training:

$$\mathbf{K}_c = C(\mathbf{K}), \quad \mathbf{V}_c = C(\mathbf{V}). \quad (6)$$

Additionally, less relevant entries are removed from the dictionary based on a relevance score ρ :

$$(\mathbf{K}_e, \mathbf{V}_e) = \text{TopK}(\rho(\mathbf{K}_i), \mathbf{K}, \mathbf{V}). \quad (7)$$

To improve retrieval robustness, we adopt a multi-query method. For each input x , we generate an augmented version of the input x' (e.g., via rotation or cropping) and extract its feature $\mathbf{Q}' = f_\theta(x')$. Both $\mathbf{Q}$ and $\mathbf{Q}'$ query the dictionary independently, and their results are combined to form the final set of reference images X_L .

B. Web-based Image Retrieval

To overcome the fundamental limitations of fixed reference dictionaries, we develop an adaptive web-retrieval framework that uses the vast amount of images available online. Fig. 2(b) illustrates our second method to generate correlated reference images: retrieving semantically relevant images with web-based image search. For each input image x , we use the Crawler tool provided by the image search engine, such as Google and Baidu, to find the top M matches

$$\mathcal{X}_W^M = \mathcal{W}(\mathbf{q}_x, M), \quad (8)$$

To maintain encoder-decoder consistency with low signaling overhead, we transmit the canonical URLs of the selected reference images. In the reported setup, only a candidate verified to be reproducible at both endpoints is admitted to

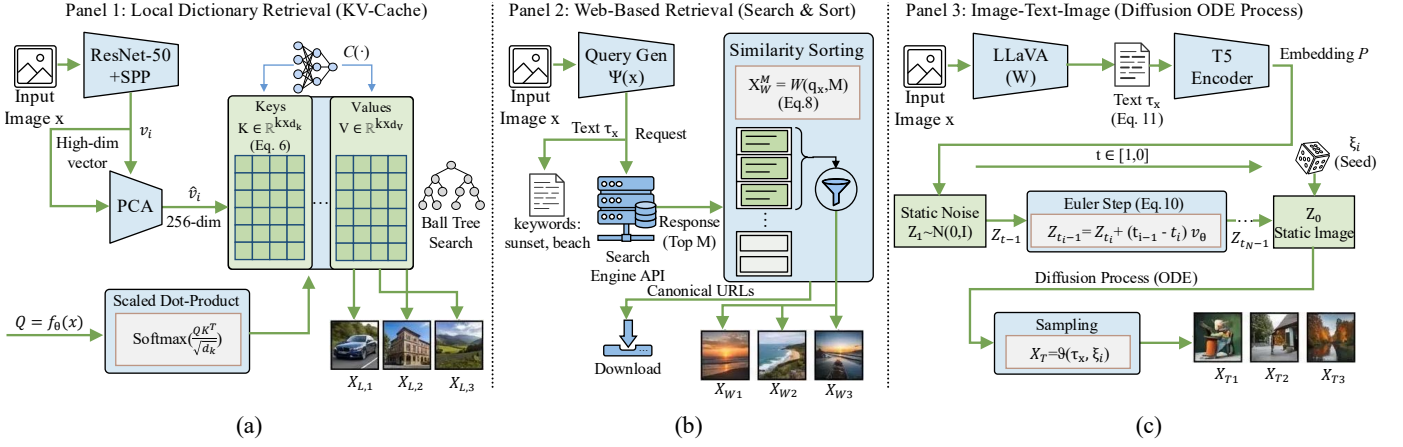


Fig. 2: Three methods (local dictionary, web search, and image-text-image generation) to obtain reference images.

final entropy coding; otherwise, the system selects the pre-synchronized local-dictionary fallback before encoding the bitstream. The resulting signaling overhead is approximately 30–100 bytes per input image.

C. Generative Image-Text-Image Synthesis

Fig. 2(c) illustrates our third method for generating correlated reference images. Our major idea is to convert the input image into text at the encoder side using a multimodal large language model, transmit the text to the decoder, and generate correlated reference images at the decoder side using text-to-image generation models. Our approach explores the potential of generative AI for efficient reference image generation. The main advantage is that only the image description is sent from the encoder to the decoder at a very low bit rate.

Specifically, for an input image x , we first generate a contextually rich semantic description $\tau_x = \mathcal{V}(x)$ using a multimodal large language model (e.g., LLaVA [29]). Then, we use a state-of-the-art diffusion-transformer-based (e.g., FLUX [30]) conditional diffusion model to generate multiple reference images. The image generation process of DiT-based diffusion models follows a rectified flow (RF), which transits the standard Gaussian noise distribution π_1 to real data distribution π_0 along a straight path. The transition between these two distributions is represented as an ordinary differential equation (ODE) over a continuous time interval $t \in [0, 1]$:

$$dZ_t = v(Z_t, t)dt, \quad (9)$$

where $Z_0 \sim \pi_0$ is the image latent representation and $Z_1 \sim \pi_1$ is the noise latent sampled from the standard Gaussian distribution $\mathcal{N}(0, I)$. During the generation process, the ODE is discretized and solved using the Euler method. Specifically, the rectified flow starts with a Gaussian noise sample $Z_{t_N} \in \mathcal{N}(0, I)$, and denoising N steps progressively along a series of discrete timesteps $t = \{t_N, \dots, t_0\}$ until generating the image latent representation Z_{t_0} :

$$Z_{t_{i-1}} = Z_{t_i} + (t_{i-1} - t_i)v_\theta(Z_{t_i}, \mathcal{P}, t_i), \quad (10)$$

where $v_\theta(\cdot)$ is the velocity prediction by the diffusion transformer and $\mathcal{P}$ is the conditional prompt embedding extracted

by the T5 text encoder [31], i.e., $\mathcal{P} = \text{T5}_{\text{text}}(\tau)$. To simplify the notation, we use $\mathcal{G}(\cdot)$ to denote the generation process of the DiT-based diffusion model. Then, our image-text-image reference generation pipeline can be represented by

$$\mathcal{X}_{\mathcal{T}}^M = \{\mathcal{G}(\tau_x, \xi_i) \mid i \in \{1, 2, \dots, M\}\}. \quad (11)$$

Sampling parameters ξ_i are derived deterministically from a cryptographic hash function. Together with matched model weights, solver settings, and software versions, they support reproducible reference generation in the reported environment. This approach offers extremely low transmission overhead and particularly excels for images with strong semantic content that can be accurately conveyed through word descriptions. We note that the “low overhead” here refers specifically to the communication bitrate (40–90 bytes per image, or less than 0.005 bpp); the computational cost of diffusion sampling is analyzed separately in Section VI.

An alternative approach would be to transmit quantized image embeddings (e.g., CLIP or DINO features) rather than text descriptions. While such embeddings can preserve richer semantic information, they incur significantly higher overhead: quantized CLIP embeddings require approximately 512 bytes compared to 40–90 bytes for text descriptions. Our text-based approach offers three advantages: substantially lower bitrate overhead, compatibility with mature and widely available text-to-image diffusion models, and interpretability through human-readable descriptions. Recent advances in image-embedding conditioned diffusion [25], [26], [27], [28] represent a promising direction for future work.

D. Adaptive Selection of Reference Method

While each reference generation method offers unique advantages, their effectiveness varies depending on the content characteristics of the input image. To maximize the compression efficiency, we implement an adaptive selection mechanism that chooses the optimal reference method for each input image, as presented in Algorithm 1. The selection

Algorithm 1: Selection of Correlated Reference Images

Input : Image x , Local dictionary D
Output: Reference images $X_r^{1,2,3}$, Selected method s^*

```

1 Function ReferenceAcquisition( $x, D$ ):
2 // Generate references using all three methods
3 // 1. Local Dictionary approach
4  $v_x \leftarrow \text{SPP}(f_\theta(x))$ 
5  $\text{indices} \leftarrow \text{KVCache}(D, v_x, 3)$ 
6  $X_L \leftarrow \{\text{GetImage}(D, i) | i \in \text{indices}\}$ 
7 // 2. Web Search approach
8  $q_x \leftarrow \Psi(x)$ 
9  $X_W \leftarrow W(q_x, 3)$ 
10 // 3. Image-Text-Image approach
11  $\tau_x \leftarrow V(x)$ 
12  $X_T \leftarrow \{G(\tau_x, \xi_i) | i \in \{1, 2, 3\}\}$ 
13 // Evaluate the performance of each method
14  $J_L \leftarrow \text{EvaluateRD}(x, X_L)$  ; // RD cost
15  $J_W \leftarrow \text{EvaluateRD}(x, X_W)$ 
16  $J_T \leftarrow \text{EvaluateRD}(x, X_T)$ 
17 // Select the best method based on RD performance
18  $s^* \leftarrow \text{argmin}\{J_L, J_W, J_T\}$ 
19 // Return the references from the best method
20 if  $s^* == L$  then
21 |  $X_r^{1,2,3} \leftarrow X_L$ 
22 else
23 | if  $s^* == W$  then
24 | |  $X_r^{1,2,3} \leftarrow X_W$ 
25 | else
26 | |  $X_r^{1,2,3} \leftarrow X_T$ 
27 return  $X_r^{1,2,3}, s^*$ 

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process can be formalized as follows: for an input image x , we generate reference images using all three approaches:

$$X_L = \mathcal{D}(x), \quad (\text{Local Dictionary}) \quad (12)$$

$$X_W = \mathcal{W}(q_x, M), \quad (\text{Web Search}) \quad (13)$$

$$X_T = \{G(\tau_x, \xi_i) | i \in \{1, 2, 3\}\}, \quad (\text{Image-Text-Image}) \quad (14)$$

where $\mathcal{D}(\cdot)$ represents the local dictionary retrieval function, $\mathcal{W}(\cdot, \cdot)$ denotes the web search function, and $G(\cdot, \cdot)$ is the generative model. For each set of references, we compute a rate-distortion score that balances compression efficiency and reconstruction quality:

$$J_s = R_s + \beta \cdot D_s, \quad s \in \{L, W, T\} \quad (15)$$

where R_s represents the estimated bit rate using reference set s (computed via the entropy model described in Section IV), D_s is the corresponding distortion measure (typically MSE), and β is a Lagrangian multiplier that balances rate and distortion

(distinct from the trade-off parameter λ used in Section V). The optimal method s^* is selected as:

$$s^* = \arg \min_{s \in \{L, W, T\}} \{J_s\} \quad (16)$$

Our experimental results (Table II) demonstrate that this adaptive approach consistently outperforms each individual reference, particularly for diverse image datasets containing various content types.

IV. CONDITIONAL LATENT SYNTHESIS AND CODING

Using the above three different approaches, we can generate semantically correlated reference images for the input, which can be synchronized between the encoder and decoder with very low communication overhead. The next question is how to utilize these correlated references. We recognize that image encoding-decoding essentially operates in the spatial domain and the major difficulty in efficient image compression lies in the effective handling of spatial high-frequency components. Although these images are close to the input image, they may not be helpful to improve the coding performance of the input image, since their spatial structures and high-frequency components could deviate significantly. Therefore, the major challenge that needs to be carefully addressed here is: *how to synthesize an effective reference from these semantically correlated images, which aligns very well with the input image and achieves highly efficient conditional coding of the input image?*

To address this challenge, our proposed GRCL method learns to synthesize a conditional reference map in the latent domain. As illustrated in Fig. 3, after the semantically correlated images are generated at the encoder, the input image x and the selected references are processed through the analysis transform network g_a to obtain their latent representations y and $Y_r^{1,2,3}$, respectively.

The analysis transform network g_a employs a hybrid architecture with a combination of CNN and Swin-Transformer modules. This design is inspired by the TCM [8] (Transformer-CNN Mixture) framework. The CNN component consists of residual blocks with down-sampling layers to capture local features, while the Swin-Transformer blocks model long-range dependencies across the image. This mixed structure enables efficient feature extraction and enhances the model's ability to generate semantically rich latent representations for conditional coding.

These latent representations are then processed through our conditional synthesis pipeline, which consists of two key modules: (1) the Conditional Latent Matching (CLM) module, which establishes precise feature correspondences between input and reference latents through sophisticated matching and spatial alignment operations; and (2) the Conditional Latent Synthesis (CLS) module, which adaptively fuses these aligned features using learnable attention mechanisms to generate the optimized conditional representation y_f . The framework also incorporates a slice-based autoregressive entropy model for efficient compression of the conditional latents. Algorithm 2 describes in detail the conditional latent coding process. This representation is then entropy coded with a hyperprior. This

unified architecture allows seamless integration of references from different sources while maintaining computational efficiency. In the following, we explain these modules in detail.

A. Conditional Latent Matching

The Conditional Latent Matching (CLM) module aims to establish precise feature correspondence between the latent representation of the input image, denoted y , and those of M reference images $Y_r^M = \{y_r^1, y_r^2, \dots, y_r^M\}$. Reference images generated by the above three approaches often exhibit spatial disparities due to their variations in scale, orientation, or structural composition. To address this issue, the proposed CLM module consists of the following two important steps: **similarity-based matching** and **multi-scale deformable alignment**.

To perform similarity-based matching, we first compute a similarity matrix S , which quantifies the affinity between features of the input latent y and each reference latent $y_r \in Y_r^M$. Initially, both y and Y_r^M undergo a feature transformation via a learnable function $\phi(\cdot)$, implemented by a convolutional neural network (CNN) comprising three 3×3 convolutional layers, each augmented with batch normalization and ReLU activations. This transformation elevates the features into a richer, high-dimensional space (e.g., 512 dimensions), modeling subtle semantic connections beyond mere pixel or latent values.

The similarity between the i -th feature of y , denoted y_i , and the j -th feature of a reference latent y_r , denoted $y_{r,j}$, is derived through a softmax-normalized dot product:

$$S_{ij} = \frac{\exp(\langle \phi(y_i), \phi(y_{r,j}) \rangle / \tau)}{\sum_k \exp(\langle \phi(y_i), \phi(y_{r,k}) \rangle / \tau)}, \quad (17)$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product, τ serves as a temperature parameter (fixed at 0.07 in our experiments) modulating the distribution's sharpness. The denominator aggregates contributions across all features k in y_r .

This results in a normalized similarity score $S_{ij} \in [0, 1]$, representing soft match between each input feature and the weighted combination of reference features. The resulting feature map y_m is synthesized by aggregating these reference features, weighted by their similarity scores:

$$y_m = \sum_j S_{ij} \cdot y_{r,j}. \quad (18)$$

Although the above similarity-based matching method is able to capture the semantic similarity between the input and its reference images in the latent domain, it is not able to effectively handle spatial transformations, such as translations, rotations, or non-rigid distortion, between the input and reference images. To address this issue, our proposed CLM performs **multi-scale deformable alignment**, denoted $F_a(y, y_m; \theta_a)$, to achieve spatial alignment of latent y_a :

$$y_a = F_a(y, y_m; \theta_a), \quad (19)$$

where θ_a represents the trainable alignment network. The architecture of F_a has a cascade of **deformable convolutional layers** operating across three distinct scales (e.g., 1×1 , 3×3 ,

and 5×5 kernels). Each layer is followed by batch normalization and GELU activations. These deformable convolutions dynamically adjust their sampling offsets based on the input, enabling adaptive alignment of features across spatial dimensions. The multi-scale approach ensures (1) **Coarse alignment** at larger scales (e.g., 5×5) to address significant structural shifts; and (2) **Fine-grained alignment** at smaller scales (e.g., 1×1) to refine local details.

The process starts with channel-wise concatenating of y and y_m , which is then fed into a deformable convolution stack. The network predicts offset at each scale for repositioning the convolution kernels' sampling points, effectively warping y_m to spatially align with y . Optimized using a rate-distortion loss during training, the parameters θ_a ensure that the above latent alignment improves the compression efficiency with minimal bit rate overhead. During encoding, latents y and Y_r^M are extracted using the analysis transform network g_a , a CNN with residual blocks and down-sampling layers. The CLM module processes these latents concurrently for each reference. The spatially aligned latent map y_a is then fed into the CLS module.

This latent-domain operation is necessary even when the external image is semantically strong. A retrieved or generated image usually differs from the target in viewpoint, scale, object position, and local texture. Direct concatenation assumes coordinate-wise correspondence that does not exist, while a PixelShuffle-style rearrangement only reindexes channels and spatial positions; it neither searches for correspondences nor rejects unrelated regions. Consequently, these simple operations can inject high-energy but incorrectly located features and produce a large PSNR loss. CLM instead searches over reference locations before learning residual spatial offsets, so reference quality and alignment quality contribute separately to the final result.

B. Conditional Latent Synthesis

The Conditional Latent Synthesis (CLS) module aims to fuse the aligned reference latent y_a with the input latent y , producing an optimized conditional latent y_f for coding of the input image. Our CLS method formulates the synthesis of y_f as a conditional probability distribution:

$$p(y_f | y, y_a) = \mathcal{N}[\mu(y, y_a), \sigma^2(y, y_a)], \quad (20)$$

where $\mu(y, y_a)$ represents the mean, representing the deterministic part of the fused latent, $\sigma^2(y, y_a)$ denotes the variance, representing the uncertainty part. Both μ and σ^2 are learned by separate neural networks. The mean $\mu(y, y_a)$ is computed through an adaptive weighting mechanism:

$$\mu(y, y_a) = \alpha \odot y + (1 - \alpha) \odot y_a, \quad (21)$$

where $\alpha \in [0, 1]$ is a spatially adaptive weight map, and $\odot$ is element-wise multiplication. The same procedure is used to predict the variance $\sigma^2(y, y_a)$. The weight map α is dynamically generated by a compact neural network $F_w([y, y_a]; \theta_f)$:

$$\alpha = \sigma\{F_w([y, y_a]; \theta_f)\}, \quad (22)$$

where $[y, y_a]$ denotes the channel-wise concatenation of y and y_a , F_w comprises two 3×3 convolutional layers with

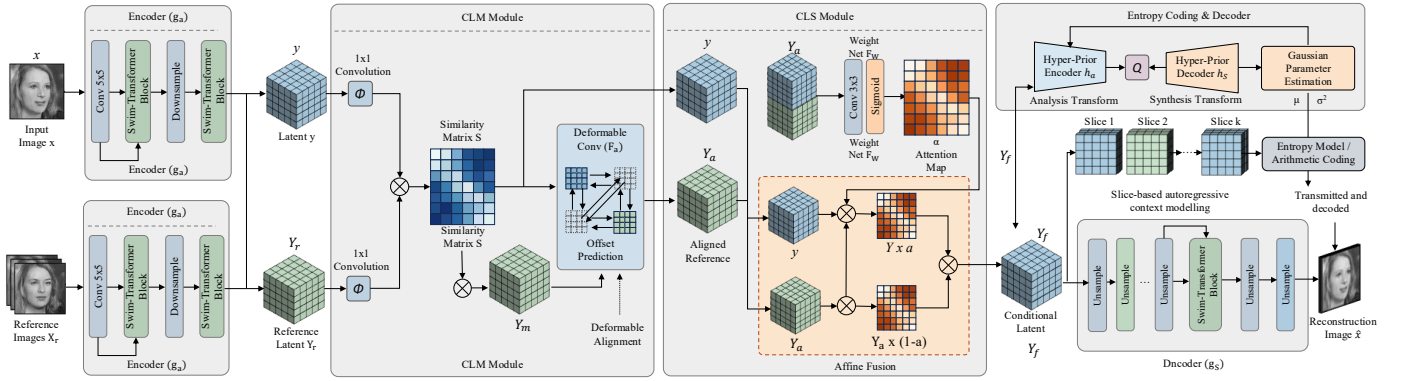


Fig. 3: Detailed architecture of our GRCL framework showing the encoding-decoding pipeline with three reference-acquisition methods, CLM, and CLS. In CLS, F_w predicts the spatial gate α , which forms the conditional latent as $\alpha \odot y + (1 - \alpha) \odot y_a$.

128 channels, succeeded by a 1×1 convolution and sigmoid activation (σ).

CLM and CLS therefore solve different failure modes. CLM determines *where* correlated reference evidence should be sampled; CLS determines *whether and how much* that aligned evidence should influence each target-latent location. If an aligned reference remains uninformative, the gate can move toward the target latent rather than forcing reference content into the code. This behavior cannot be reproduced by a fixed average or a spatial rearrangement. The same-reference ablations in Section VI keep the selected external images fixed while modifying this latent processing, thereby isolating the architectural contribution from the quality of the external side information.

C. Entropy Coding with Hyperprior

To achieve efficient compression in our GRCL framework, a hyperprior network h_a is utilized to extract a compact hyperprior z from the conditional latent representation y_f , which is derived from the input image and its references:

$$z = h_a(y_f). \quad (23)$$

The hyperprior z is quantized into $\hat{z} = Q(z)$ and encoded independently using a factorized entropy model. This quantized hyperprior provides critical contextual information, significantly enhancing the accuracy of probability estimation for the entropy coding of y_f .

For the entropy coding process, we implement a slice-based autoregressive context model. The conditional latent y_f is partitioned into K slices along the channel dimension: $y_f = [y_f^1, y_f^2, \dots, y_f^K]$. Each slice y_f^i is encoded sequentially, with its probability distribution conditioned on both the previously encoded slices $y_f^{<i} = [y_f^1, \dots, y_f^{i-1}]$ and the quantized hyperprior $\hat{z}$:

$$p(y_f^i | y_f^{<i}, \hat{z}) = f_\theta(y_f^{<i}, \hat{z}), \quad (24)$$

where f_θ is a neural network parameterized by θ . This network predicts the parameters of a Gaussian distribution for each element in y_f^i :

$$p(y_f^i | y_f^{<i}, \hat{z}) \sim \mathcal{N}(\mu_i, \sigma_i^2), \quad (25)$$

Algorithm 2: Conditional Latent Coding with Generated References

Input : Image x , Reference images $X_r^{1,2,3}$

Output: Conditional latent y_f , Encoded bitstream

- 1 $y \leftarrow g_a(x)$
- 2 $Y_r^{1,2,3} \leftarrow \{g_a(x_r^i) | i \in \{1, 2, 3\}\}$
- 3 **Conditional Latent Matching (CLM):**
- 4 **for each reference latent** $y_r \in Y_r^{1,2,3}$ **do**
- 5 $S_{ij} \leftarrow \frac{\exp(\langle \phi(y_i), \phi(y_{r,j}) \rangle / \tau)}{\sum_k \exp(\langle \phi(y_i), \phi(y_{r,k}) \rangle / \tau)}$
- 6 $y_m \leftarrow \sum_j S_{ij} \cdot y_{r,j}$
- 7 $y_a \leftarrow F_a(y, y_m; \theta_a)$ using deformable alignment
- 8 **Conditional Latent Synthesis (CLS):**
- 9 $\alpha \leftarrow \sigma(F_w([y, y_a]; \theta_f))$
- 10 $\mu(y, y_a) \leftarrow \alpha \odot y + (1 - \alpha) \odot y_a$
- 11 $y_f \sim \mathcal{N}(\mu(y, y_a), \sigma^2(y, y_a))$
- 12 **Entropy Coding:**
- 13 $z \leftarrow h_a(y_f)$
- 14 $\hat{z} \leftarrow Q(z)$
- 15 Partition y_f into K slices: $y_f = [y_f^1, y_f^2, \dots, y_f^K]$
- 16 **for** $i \leftarrow 1$ **to** K **do**
- 17 $(\mu_i, \sigma_i) \leftarrow f_\theta(y_f^{<i}, \hat{z})$
- 18 $p(y_f^i | y_f^{<i}, \hat{z}) \sim \mathcal{N}(\mu_i, \sigma_i^2)$
- 19 $\hat{y}_f^i \leftarrow Q(y_f^i)$
- 20 **return** y_f , EncodedBitstream

with $(\mu_i, \sigma_i) = f_\theta(y_f^{<i}, \hat{z})$. During encoding, each slice y_f^i is quantized to $\hat{y}_f^i$, and a residual is calculated as:

$$r_i = y_f^i - \hat{y}_f^i, \quad (26)$$

to mitigate quantization errors. These residuals can be optionally encoded to enhance reconstruction quality, particularly at lower bitrates. It can be seen that we generate the conditional reference in the latent domain based on the CLM and CLS modules. Because of this, we refer to our proposed method as generative reference of conditional latents.

Note that the reconstruction is deterministic at inference time. The Gaussian $\mathcal{N}(\mu_i, \sigma_i^2)$ parameterizes the entropy

coder; no stochastic sampling occurs. Given identical quantized latents, references, model weights, and inference settings, the output is bit-exact, consistent with standard learned compression practice.

D. Unified Decoding with Different Reference Generation Approaches

The decoding process in GRCL is engineered for flexibility and efficiency, supporting multiple reference generation methods while ensuring accurate reconstruction of the input image. The decoder mirrors the network architecture of the encoder, incorporating a hybrid structure of CNN and Swin-Transformer modules similar to the TCM approach. Specifically, the synthesis network g_s consists of a series of upsampling CNN layers interleaved with Swin-Transformer blocks to capture both local and global dependencies in the latent space. This hybrid design ensures that the decoder can effectively reconstruct high-frequency details while maintaining semantic consistency with the input image.

The process begins with the decoder extracting the quantized hyperprior $\hat{z}$ from the bitstream, which provides essential context for entropy modeling. The conditional latent representation $\hat{y}_f$ is then reconstructed slice-by-slice using the autoregressive context model, conditioned on $\hat{z}$ and previously decoded slices. After obtaining the references, the decoder processes them through a feature extraction pipeline identical to the encoder's, involving latent space mapping, spatial alignment via conditional latent matching, and feature synthesis to guide the reconstruction of $\hat{y}_f$. The synthesis network g_s then transforms $\hat{y}_f$ into the final reconstructed image $\hat{x}$, using the reference route selected at the encoder and signaled in the bitstream.

Encoder–decoder synchronization: The local dictionary (Method 1) is pre-synchronized and self-contained. For web search (Method 2), the reported setup uses canonical URLs; for image-text-image generation (Method 3), fixed random seeds and specified model/software versions reproduce the references in the reported environment. A pre-coding consistency check admits only a reference that can be reproduced identically at both endpoints. If this check fails, the system selects the pre-synchronized local-dictionary branch before encoding the bitstream. Thus, the local-dictionary branch is the deterministic offline option, whereas the web and generative branches require the stated external resources.

V. GENERALIZATION ANALYSIS OF REFERENCE PERTURBATIONS

This section studies a specific property of reference-conditioned coding: how the recoverability of a shared latent structure changes as a reference becomes less correlated with the target. We use a spiked-covariance abstraction in which the target and reference share a low-rank component and the reference additionally contains unrelated variation. The analysis is directional rather than module-by-module: it relates reference correlation and cross-covariance estimation error to shared-subspace recovery.

The abstraction does not model the full GRCL pipeline. The variables z and $\hat{z}$ are abstract target and reference observations rather than literal network tensors; their shared low-rank structure is only a proxy for useful conditional structure after feature extraction and alignment. The correlation γ qualitatively corresponds to post-matching target–reference correlation, and the perturbation term represents uncorrelated or mismatched reference content. Subspace recovery is not equal to PSNR or BD-rate. The analysis does not characterize search-engine ranking, caption or diffusion errors, duplicate retrieval, the exact deformable-convolution architecture, or interactions with slice-wise autoregressive entropy coding, and it does not establish that the chosen CLM/CLS implementation is optimal. Instead, it motivates the design objective of preserving correlated reference structure after alignment and attenuating uncorrelated components during fusion. Empirical ablations are required to assess each implemented module.

A. Problem Formulation

Let $z \in \mathbb{R}^d$ denote the original image and $\hat{z} \in \mathbb{R}^d$ the reference image serving as side information. We define encoders $E_1 : \mathbb{R}^d \rightarrow \mathbb{R}^k$ and $E_2 : \mathbb{R}^d \rightarrow \mathbb{R}^k$ for the original and reference images, respectively, and a decoder $F : \mathbb{R}^k \times \mathbb{R}^k \rightarrow \mathbb{R}^d$. The rate-distortion optimization problem is formulated as:

$$\min_{E_1, E_2, F} \mathbb{E}_{z, \hat{z}} [R(E_1(z), E_2(\hat{z})) + \lambda \cdot D(z, F(E_1(z), E_2(\hat{z})))], \quad (27)$$

where $R(\cdot, \cdot)$ represents the rate (compression) loss, $D(\cdot, \cdot)$ denotes the distortion loss (e.g., reconstruction error), and $\lambda > 0$ is a parameter balancing rate and distortion.

B. Assumptions

We introduce the following assumptions:

Assumption 1. Spiked Covariance Model for Images - The original image z follows a spiked covariance model:

$$z = V^* \alpha + \eta, \quad (28)$$

where $V^* \in \mathbb{R}^{d \times k}$ is the true low-rank feature matrix with orthonormal columns ($V^{*T} V^* = I_k$), $\alpha \in \mathbb{R}^k$ is the latent representation with $\mathbb{E}[\alpha] = 0$ and covariance $\mathbb{E}[\alpha \alpha^T] = \Omega_\alpha$, and $\eta \in \mathbb{R}^d$ is additive noise, independent of α , with zero mean and covariance $\Omega_\eta = \sigma_\eta^2 I_d$.

Assumption 2. Reference Image with Irrelevant Parts - The reference image $\hat{z}$ is modeled as:

$$\hat{z} = V^* (\gamma \alpha + \sqrt{1 - \gamma^2} \alpha_\perp) + \hat{\eta}, \quad (29)$$

where $\gamma \in [0, 1]$ quantifies the correlation between z and $\hat{z}$, $\alpha_\perp \in \mathbb{R}^k$ is independent of α with $\mathbb{E}[\alpha_\perp] = 0$ and $\mathbb{E}[\alpha_\perp \alpha_\perp^T] = \Omega_\alpha$, and $\hat{\eta} \in \mathbb{R}^d$ is additive noise, independent of α and $\alpha_\perp$, with zero mean and covariance $\Omega_{\hat{\eta}} = \sigma_{\hat{\eta}}^2 I_d$. The proportion of irrelevant information in $\hat{z}$ is defined as $q = 1 - \gamma^2$.

Assumption 3. Entropy Model for Rate Loss - The rate loss follows a Gaussian entropy model:

$$R(y, \hat{y}) = \mathbb{E}_{y, \hat{y}} [-\log_2 p_\phi(y | \hat{y})], \quad (30)$$

where $p_\phi(y | \hat{y}) = \mathcal{N}(y; \mu(\hat{y}), \Omega_y)$ is a conditional Gaussian distribution with mean $\mu(\hat{y})$ and covariance Ω_y conditioned on $\hat{y}$.

Assumption 4. Distortion Loss - The distortion loss is the mean squared error between the original and reconstructed images:

$$D(z, \tilde{z}) = \|z - \tilde{z}\|_2^2, \quad (31)$$

where $\tilde{z} = F(E_1(z), E_2(\hat{z}))$.

Assumption 5. Sub-Gaussian Noise - The noise vectors η and $\hat{\eta}$ are sub-Gaussian with parameter $\sigma^2 = \sigma_\eta^2$, satisfying:

$$\begin{aligned} \mathbb{P}(|u^T \eta| \geq t) &\leq 2 \exp\left(-\frac{t^2}{2\sigma^2}\right), \\ \forall t > 0, \quad \forall u \in \mathbb{R}^d, |u|_2 &= 1. \end{aligned} \quad (32)$$

C. Main Results

Lemma 1. Under Assumption 3, consider the ideal matched-model case in which $y \in \mathbb{R}^k$, $\Omega_y \succ 0$ is independent of $\hat{y}$, and the true conditional density satisfies $q(y | \hat{y}) = p_\phi(y | \hat{y})$ almost surely. Then the rate loss is given by:

$$R(y, \hat{y}) = \frac{1}{2 \ln 2} (k \ln(2\pi e) + \ln \det(\Omega_y)). \quad (33)$$

Proof. Under the matched-model condition, $R(y, \hat{y})$ equals the conditional differential entropy $h(y | \hat{y})$ converted from nats to bits. Since each conditional Gaussian has covariance Ω_y independent of $\hat{y}$,

$$h(y | \hat{y}) = \frac{1}{2} \ln((2\pi e)^k \det(\Omega_y)). \quad (34)$$

Therefore, $R(y, \hat{y}) = h(y | \hat{y}) / \ln 2$, which yields Equation (33). $\square$

Lemma 1 is an ideal matched-Gaussian rate identity rather than an end-to-end rate guarantee. Assumptions 4–5 fix the distortion measure and noise regime of the rate-distortion setup, while the recovery statement below depends only on Assumptions 1 and 2.

The following proposition formalizes the directional role of target-reference correlation in the simplified model:

Proposition 1 (Directional Effect of Reference Correlation). Under Assumptions 1 and 2, assume that the target and reference noise terms are mutually independent and independent of the latent coefficients. For m paired samples, define the empirical target-reference cross-covariance

$$\hat{C}_{z\hat{z}} = \frac{1}{m} \sum_{i=1}^m z_i \hat{z}_i^T, \quad (35)$$

and let the columns of $\hat{V} \in \mathbb{R}^{d \times k}$ be its leading k left singular vectors. If $\|\hat{C}_{z\hat{z}} - C_{z\hat{z}}\|_2 \leq \epsilon < \gamma \lambda_{\min}(\Omega_\alpha)$, then

$$\left\| \sin \Theta(\text{span}(\hat{V}), \text{span}(V^*)) \right\|_F \leq \frac{\sqrt{k} \epsilon}{\gamma \lambda_{\min}(\Omega_\alpha) - \epsilon}, \quad (36)$$

where $C_{z\hat{z}} = \mathbb{E}[z \hat{z}^T]$.

Proof. Using the independence and zero-mean assumptions in the proposition,

$$C_{z\hat{z}} = \mathbb{E}[z \hat{z}^T] = \gamma V^* \Omega_\alpha V^{*T}. \quad (37)$$

Therefore, $C_{z\hat{z}}$ has rank k , its leading left singular subspace is $\text{span}(V^*)$, and its smallest nonzero singular value is $\gamma \lambda_{\min}(\Omega_\alpha)$. Applying the standard singular-subspace perturbation bound to $\hat{C}_{z\hat{z}} = C_{z\hat{z}} + (\hat{C}_{z\hat{z}} - C_{z\hat{z}})$ gives the stated result whenever the estimation error ϵ is smaller than this singular-value gap. $\square$

Proposition 1 is intentionally directional. For a fixed cross-covariance estimation error ϵ , stronger target-reference correlation γ enlarges the shared-subspace gap and tightens the perturbation bound; as $\gamma \rightarrow 0$, the proposition provides no guarantee. It analyzes a shared-subspace estimator rather than the trained GRCL codec and therefore does not assign a quantitative gain to CLM, CLS, or any other module. Those effects are evaluated by the component ablation in Table VI, the same-reference controls in Table VII, and the controlled reference perturbations in Fig. 7.

VI. EXPERIMENTAL RESULTS

In this section, we present extensive experiments to evaluate our proposed GRCL method. We first describe the experimental setup including datasets, implementation details, and evaluation metrics. Then, we compare our method with state-of-the-art image compression approaches. Finally, we provide comprehensive ablation studies to further understand our method.

A. Datasets

The following three datasets are used in our method: **(1) Training:** The Flickr2W dataset [32] with 20,745 high-quality images is used for training. **(2) Dictionary Construction:** The Flickr2K dataset [33] with 2,650 images, which were randomly cropped into 256×256 patches, is used as our reference dictionary.

We conducted experiments on public benchmark datasets and a custom-created private dataset that provides a leakage-resistant supplementary evaluation on privately collected natural scenes. We used two widely used public datasets for performance evaluation and comparison with existing methods: **(1) Kodak Dataset** [34]: a standard test set in image compression research consisting of 24 uncompressed color images of 768×512 resolution. This dataset features a diverse range of natural scenes with different levels of details and texture complexity. **(2) CLIC Professional Validation Set** [35]: The Challenge on Learned Image Compression (CLIC) professional validation set consists of 41 high-resolution images. These images provide a challenging benchmark for image compression.

To reduce the possibility that a public test image has appeared in a retrieval index or a generative-model corpus, we additionally evaluate on the Private High-Resolution Dataset (PHRD). The original PHRD contains 32 privately collected high-resolution images whose longest side is 1024 pixels

and that primarily depict natural outdoor scenery. We expand PHRD to 96 images by adding 64 privately collected images from the same natural-scene domain. Both subsets primarily depict natural outdoor scenery and contain no indoor scenes or portraits. None of the PHRD targets is included in Flickr2W, the Flickr2K reference dictionary, or model training. Before evaluation, retrieved candidates are screened for exact copies, crops, and near duplicates with a conservative MinHash stage, followed by perceptual-hash and feature-similarity checks and manual inspection; a rejected candidate is replaced by the next ranked result. No duplicate target–reference pair was found for the original or expanded PHRD under this protocol.

The expanded 96-image PHRD is larger than Kodak (24) and CLIC Professional validation (41) and provides an additional leakage-resistant check. We report the original 32-image PHRD and the expanded 96-image set separately, and avoid using the private set as the sole basis for a general fairness claim. On the original/expanded PHRD, GRCL achieves -15.2% / -15.1% BD-Rate_P relative to VTM; the expanded-set result remains -15.1% after duplicate screening. Rate-distortion curves in Fig. 4 include the expanded 96-image PHRD panel.

B. Experimental Settings

Our model was implemented in PyTorch and trained on 8 NVIDIA A100 GPUs. The training process had 30 epochs using the Adam optimizer with an initial learning rate of 1×10^{-4} , reduced by a factor of 0.5 every 10 epochs. We used a batch size of 16 per GPU and a patch size of 16×16 pixels for the matching module. The slice-based autoregressive context model used 5 slices for entropy coding.

When generating the reference images using those three approaches, the experimental settings are: **(1) Local Dictionary:** We used a dictionary with 3,000 clusters. The number was determined empirically to balance performance and computational efficiency. **(2) Web Search:** We implemented a Baidu API-based image retrieval system with a 0.5-second response time. Retrieved images were resized to match our training patch dimensions (256×256) to ensure compatibility with the network architecture. **(3) Image-Text-Image Generation:** We used the LLaVA [20] model for generating text descriptions of the input images. We used the FLUX method to generate reference images based on these textual descriptions. Generated images were sized to match training patch dimensions (256×256) during training. During testing, we dynamically generated reference images matching the dimensions of the test images to ensure optimal spatial alignment for conditional coding.

Unless otherwise specified, every selected branch supplies three references. For comparisons among fusion architectures, the acquired reference images, their order, and their signaling bits are held fixed; only the operation that maps those references into a conditional latent is changed. This same-reference, same-overhead protocol prevents a stronger retrieval result from being attributed to CLM or CLS. Near-duplicate filtering (conservative MinHash, then perceptual-hash / feature checks, followed by manual inspection) is applied to the top-three web-retrieval candidates for each image before adaptive

TABLE I: BD-rate savings (%) compared to VTM on Kodak and CLIC datasets. BD-Rate_P refers to PSNR-based BD-rate, while BD-Rate_M refers to MS-SSIM-based BD-rate.

Method	Kodak		CLIC	
	BD-Rate _P	BD-Rate _M	BD-Rate _P	BD-Rate _M
BPG [36]	+2.3	+3.5	+2.1	+3.2
AV1 [41]	-5.2	-6.1	-4.9	-5.8
Hyperprior [5]	-7.1	-8.3	-6.7	-7.9
Cheng et al. [7]	-9.2	-10.4	-8.7	-9.9
ELIC [40]	-10.5	-11.9	-10.1	-11.3
TCM [8]	-12.7	-13.4	-12.1	-12.8
CLC [3]	-14.5	-15.2	-13.9	-14.7
GRCL (Ours)	-16.0	-16.7	-15.4	-16.1

branch selection, so rejected candidates cannot influence either the selected branch or the reported rate-distortion result. Reject rates are counted over these web-retrieval candidates, not over images or candidates from the other acquisition routes.

We evaluate our method using rate-distortion (RD) performance, measured with PSNR and MS-SSIM metrics. We also compute the BD-rate savings relative to VTM, representing the average bitrate reduction while maintaining the same quality level.

C. Comparison with State-of-the-Art Methods

We conducted extensive experiments to evaluate the rate-distortion performance of our proposed GRCL method against both traditional codecs including BPG [36], VTM [37], HM [38], and JPEG [39], as well as recent learning-based methods including the Hyperprior [5], Cheng et al. [7], ELIC [40], TCM [8], and our preliminary conference version CLC [3] methods. We also include results from the AV1 method [41] for performance comparison.

Figure 4 shows the rate-distortion curves for Kodak, CLIC, and the expanded PHRD (96 images). GRCL outperforms the compared codecs across the plotted bit-rate range on all three sets, retaining a slight lead at high rates. The improvement is most significant at low bit-rates, which are challenging for conventional compression techniques. The aggregate improvements over VTM are summarized by the BD-rate results in Table I.

Table I summarizes the BD-rate savings when compared to the VTM method, which was used as the baseline in the compression research community. Our GRCL method achieves the highest BD-rate savings among all methods, with -16.0% and -16.7% on the Kodak dataset for PSNR and MS-SSIM metrics, respectively. Similar performance improvement was observed on the CLIC dataset, with -15.4% and -16.1% BD-rate savings for PSNR and MS-SSIM, respectively. On Kodak, GRCL provides an additional 1.5 percentage points of BD-rate saving over our preliminary CLC method for both metrics, supporting the contribution of generative references.

Figure 5 provides a qualitative comparison with visual examples of reconstructed images. Our GRCL method demonstrates superior fidelity in preserving textural details and structural elements, particularly at challenging low bit rates. The zoomed-in regions (highlighted by red and blue boxes) reveal

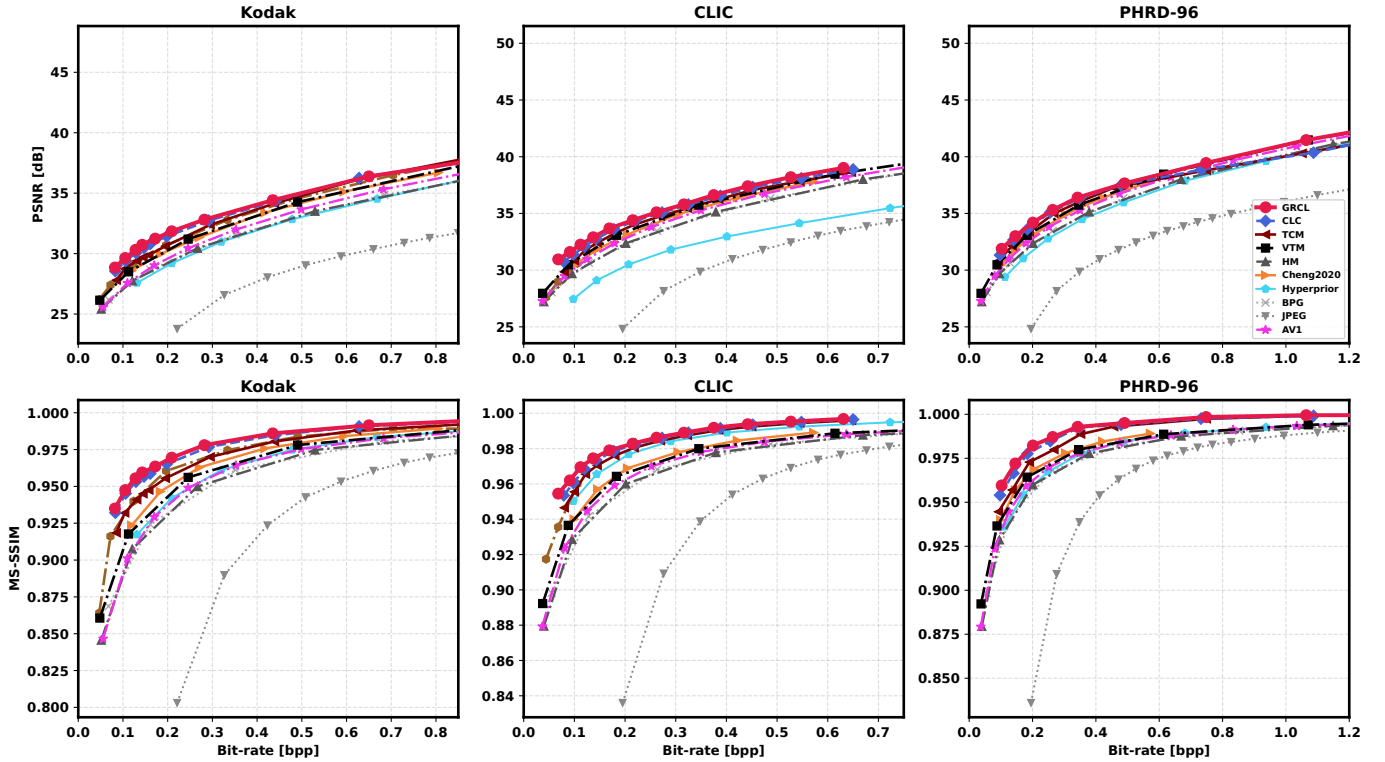


Fig. 4: Rate-distortion performance comparison on the Kodak, CLIC, and expanded PHRD (96 images) datasets, evaluated with PSNR (top) and MS-SSIM (bottom). The proposed GRCL outperforms traditional codecs (VTM, HM, BPG, AV1, JPEG) and recent learned codecs (Hyperprior, Cheng et al., ELIC, TCM, CLC) across the plotted bit-rates on all three datasets, with the largest margins at low rates.

that GRCL successfully maintains fine-grained details such as building textures, railings, and natural landscape features that are typically compromised in existing methods.

D. Ablation Studies

In this section, we provide extensive ablation studies to further understand our method.

1) *Evaluation of Reference Generation Methods*: We first evaluate the performance of each of these three methods for reference generation. Table II shows the BD-rate savings compared to VTM when using different reference generation methods. The results show that each method provides significant improvement over the baseline (no reference). Web search achieves the best performance among individual methods, followed by local dictionary and image-text-image generation. However, our adaptive selection mechanism outperforms all individual methods, demonstrating the advantage of dynamically selecting the optimal reference method for each input image.

Figure 6 shows the distribution of different generation methods for the conditional references across different image categories in the Kodak dataset. Web search is mostly chosen for images with distinctive landmarks or common objects, while image-text-image generation is favored for images with strong semantic elements that can be effectively captured in text. The local dictionary method is mainly chosen for images with subtle textures or common patterns.

TABLE II: BD-rate savings (%) vs. VTM for different reference acquisition methods on Kodak dataset.

Reference Method	BD-Rate _P	BD-Rate _M
No Reference (Baseline)	-12.7	-13.4
Local Dictionary Only	-14.5	-15.2
Web Search Only	-15.1	-15.8
Image-Text-Image Only	-14.2	-14.9
Adaptive Selection	-16.0	-16.7

2) *Analysis of Bit Rate Overhead for the Reference Information*: For these three generation methods, the reference identifiers or generation parameters must be transmitted to support encoder-decoder consistency. We analyze the overhead associated with each reference generation method in Table III. Despite the additional information required for reference signaling, the overall impact on the bitrate remains small relative to the payload (on the order of one percent at 0.2 bpp on Kodak 768×512). As shown in Figure 6, the local dictionary approach requires the fewest signaling bits ($\approx \log_2 K$ bits per index; with $K=3000$ and three references this is about 36 bits) compared to web search (hundreds of bits for URLs) and image-text-image captions (hundreds of bits). While the local dictionary can capture some texture information with minimal overhead, web search can return an exact copy or crop among its initial candidates, but such candidates are rejected by the duplicate-screening protocol

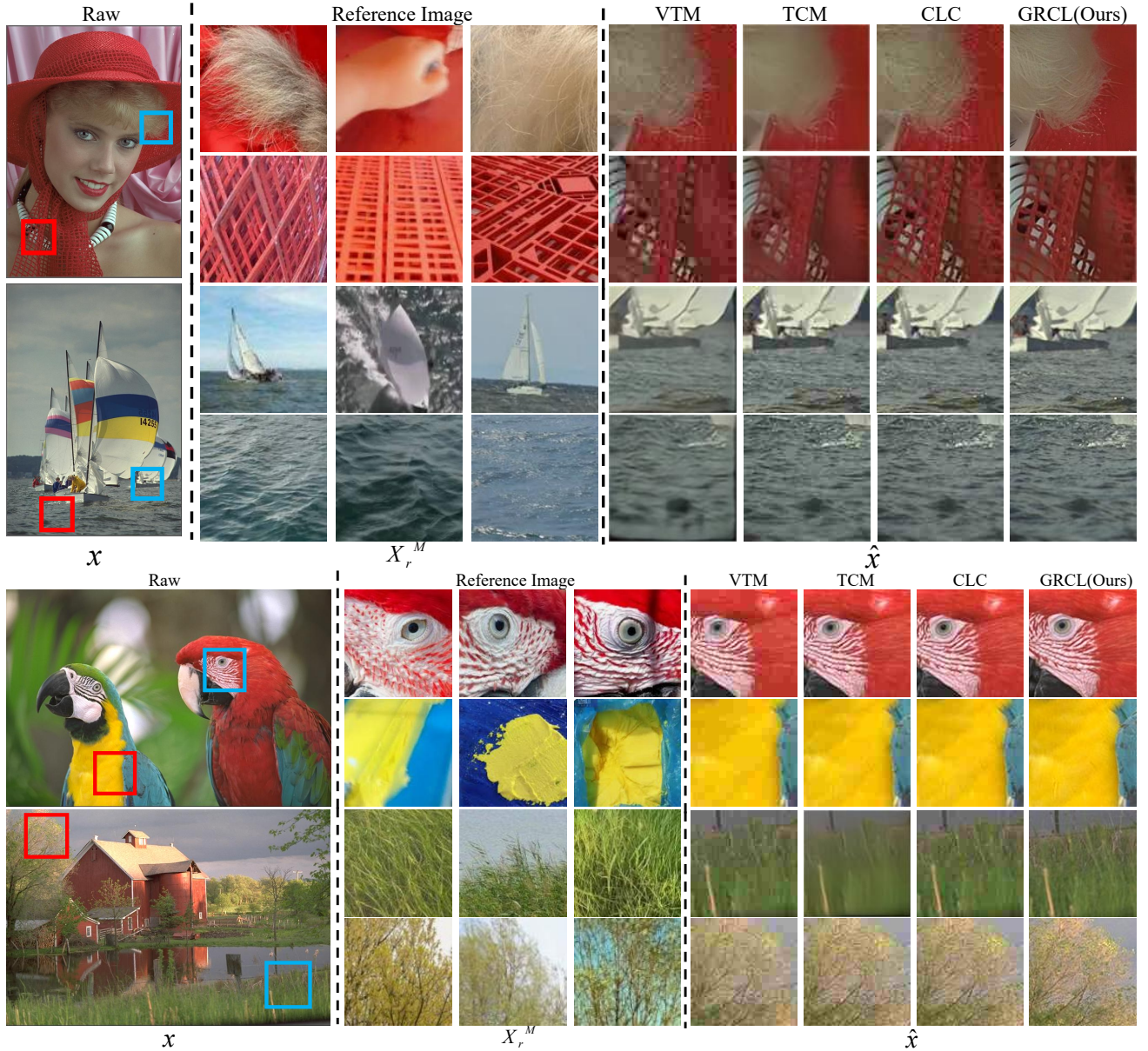


Fig. 5: Visual comparison of reconstruction quality across different compression methods. The left columns show the original image and our synthesized reference, while subsequent columns compare reconstructions from VTM, TCM, CLC, and our GRCL method. The red and blue boxes highlight magnified regions that demonstrate GRCL’s superior preservation of fine textures and structural details at comparable bit rates.

before adaptive branch selection and are not used for coding. Valid web references require more signaling bits than local-dictionary indices. The primary advantage of generative methods lies in their resolution alignment capabilities, providing better adaptability to the target compression requirements. The results confirm that our reference signaling mechanism incurs minimal overhead while providing substantial gains in compression performance.

3) *Performance Analysis Across Different Image Types*: To better understand the effectiveness of our adaptive selection mechanism, we manually categorize the 24 Kodak images into five content types based on their dominant visual characteristics: *Natural Scenes* (landscapes, outdoor environments

TABLE III: Average reference overhead (in bytes and as a percentage of total bitrate) at 0.2 bpp on Kodak dataset.

Reference Method	Bytes	% of Total	BD-Rate Impact
Local Dictionary Indices	4-8	0.04-0.08%	0.05%
Web Search URLs	30-100	0.31-1.02%	0.15%
Text Captions	40-90	0.41-0.92%	0.10%
Total (Average)	76	0.77%	0.31%

with natural elements), *Portraits* (human faces and figures), *Architecture* (buildings, man-made structures), *Objects* (close-up shots of specific items), and *Textures* (images dominated by repetitive patterns or surface details). This categorization

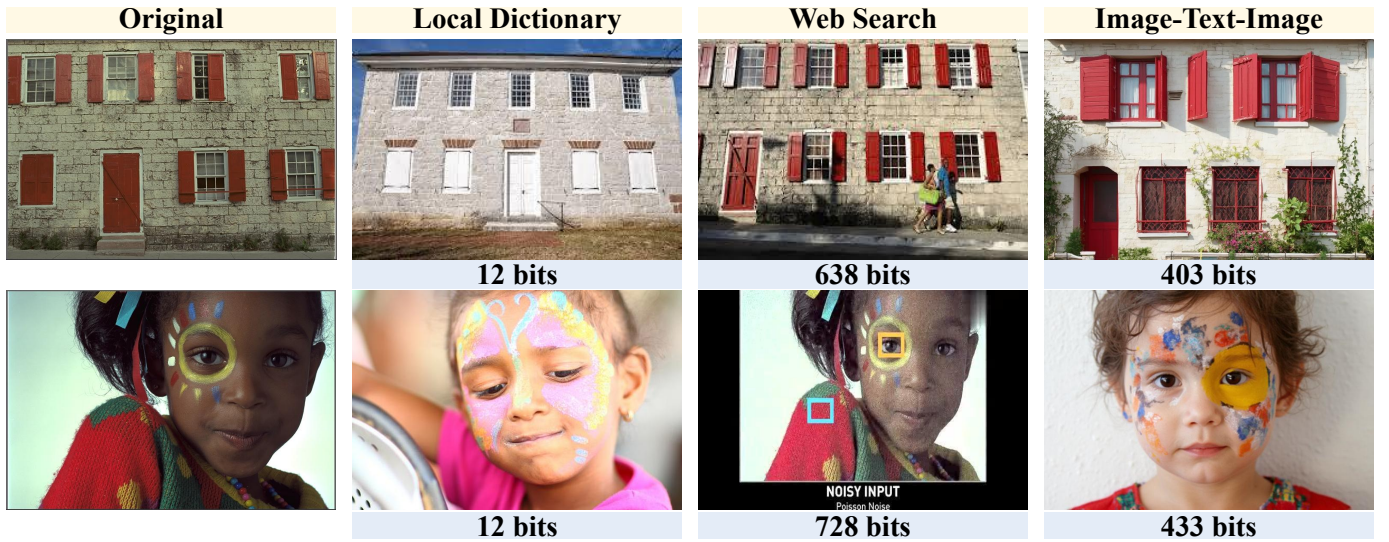


Fig. 6: Comparison of bit overhead across different reference methods. Local dictionary uses compact indices ($\approx \log_2 K$ bits each), while web search and image-text-image methods consume substantially more bits for URLs or captions, illustrating the signaling trade-offs among the three routes.

TABLE IV: BD-rate savings (%) vs. VTM for different image content types using the best-performing reference method on Kodak dataset.

Image Type	Count	Best Method	BD-Rate _P (%)
Natural Scenes	8	Web Search	-17.2
Portraits	6	Image-Text-Image	-15.8
Architecture	5	Web Search	-16.5
Objects	3	Local Dictionary	-14.9
Textures	2	Local Dictionary	-13.7
Overall (Adaptive)	24	Adaptive Selection	-16.0

is performed by visual inspection to analyze how different content types benefit from our reference generation methods.

Table IV reveals interesting patterns in method selection effectiveness across different content types. Natural scenes achieve the highest compression gains (-17.2% BD-rate) using web search, benefiting from the abundance of similar landscape images in online databases. Portrait images favor the image-text-image approach (-15.8% BD-rate), as semantic descriptions effectively capture facial features, poses, and human characteristics that are well-represented in text-to-image generation models. Architecture images also perform well with web search (-16.5% BD-rate), leveraging common building structures and architectural patterns available online. Object and texture images show preference for the local dictionary approach, likely due to their reliance on fine-grained visual patterns that are better captured through direct feature matching rather than semantic description. The adaptive selection mechanism successfully identifies the optimal method for each content type, achieving overall performance that balances these individual strengths.

Across Kodak, CLIC, and expanded PHRD, the final local/web/generated selection counts are 5/13/6, 8/23/10, and 28/40/28, respectively. The corresponding aggregate fallback counts are 1, 2, and 3. This distribution shows that the reported

TABLE V: BD-rate savings (%) vs. VTM for different numbers of reference images.

Num of Refs	Kodak		CLIC	
	BD-Rate _P	BD-Rate _M	BD-Rate _P	BD-Rate _M
1	-13.8	-14.5	-13.2	-13.9
2	-15.3	-16.0	-14.7	-15.4
3	-16.0	-16.7	-15.4	-16.1
4	-15.9	-16.6	-15.3	-16.0
5	-15.8	-16.5	-15.2	-15.9

result is not dominated by a single acquisition route. An aggregate fallback is recorded when the initially selected candidate fails the pre-coding availability or endpoint-consistency check, causing the system to advance to another feasible route or the pre-synchronized local fallback before encoding the bitstream. Because the available log records fallback at the system level, these counts are not presented as separate per-branch failure rates.

4) *Ablation on Number of Reference Images*: Table V shows the impact of the number of reference images (ranging from 1 to 5) on the compression performance. We can see that using three reference images achieves the best performance on both datasets, providing BD-rate savings of 16.0% and 15.4% on Kodak and CLIC, respectively. Having more than three images introduces redundancy and leads to slightly degraded performance, confirming our design choice of using three reference images as the default setting.

5) *Contribution of Different Algorithm Components*: Table VI separates reference acquisition from latent processing. Removing web search or image-text-image generation changes the available side information and reduces PSNR-based BD-rate savings by 1.5 and 0.9 percentage points, respectively. In contrast, the w/o-CLM and w/o-CLS variants retain the selected references but bypass the corresponding latent operation; their savings fall by 2.3 and 4.8 percentage points.

TABLE VI: BD-rate savings (%) vs. VTM for different model configurations on Kodak. External references are held fixed in the CLM/CLS rows.

Model Configuration	BD-Rate _P	BD-Rate _M	Encoding Time (s)
Full Model	-16.0	-16.7	0.88
w/o Web Search	-14.5	-15.2	0.95
w/o Image-Text-Image	-15.1	-15.8	0.90
w/o CLM	-13.7	-14.4	0.88
w/o CLS	-11.2	-12.0	0.83
w/o KV-cache	-15.9	-16.6	1.75

TABLE VII: Same-reference controls on Kodak. All fusion variants use the reference set selected for GRCL; the no-reference codec uses none and is set to 0.0 percentage points (pp). Each entry is the difference between the method's VTM-relative BD-rate and that of the no-reference codec, not a re-normalized relative percentage. More negative is better.

Variant	BD-Rate _P change (pp)	BD-Rate _M change (pp)
No-reference codec	0.0	0.0
Direct concatenation	+1.7	+1.8
PixelShuffle rearrangement	+2.4	+2.5
Fixed average fusion	+1.3	+1.4
Identity alignment + simple fusion	+1.1	+1.2
Global affine alignment + simple fusion	-0.1	0.0
Simple attention fusion	-0.4	-0.4
w/o CLM	-1.0	-1.0
w/o CLS	+1.5	+1.4
Full GRCL	-3.3	-3.3

Thus, access to correlated external images is beneficial but does not by itself account for the full gain. Table VII reports percentage-point changes in VTM-relative BD-rate after setting the no-reference codec to 0.0 pp. Simple same-reference fusions remain near or worse than this baseline, whereas full GRCL provides 3.3 pp more BD-rate saving under the same references.

E. Computational Complexity Analysis

To evaluate computational cost, we separate reference acquisition from neural encoding and decoding under identical model weights and settings on Kodak (768×512). Total latency is Ref.+Enc.+Dec., and network-dependent web latency is treated as acquisition time. The A100 totals for the baseline, local-dictionary, web-search, and image-text-image routes are 1.47, 1.58, 2.03, and 7.53–9.53 s, respectively; the corresponding RTX 3090 totals are 1.88, 2.02, 2.46, and 9.96–12.96 s. Peak memory ranges from 3.1 to 17.8 GB on A100 and from 3.3 to 20.8 GB on RTX 3090.

These values are measured route by route. They do not include a single wall-clock measurement that generates and evaluates all three external routes before adaptive selection. On A100, the local-dictionary branch adds only 0.05 s of reference acquisition and is the appropriate option for latency-sensitive or offline deployment. Web retrieval adds approximately 0.50 s and depends on network and service availability. Image-text-image generation dominates its route latency at 6–8 s, even though neural codec time remains 0.88 s for encoding and 0.65 s for decoding. Consequently, GRCL should

be viewed as a configurable rate-distortion-latency system: local retrieval serves lower-latency settings, whereas web and generative branches are intended for bandwidth-constrained, delay-tolerant use.

F. Generalization Experiments

To test the robustness trend predicted by the simplified analysis, we perturb the reference input during inference. The stress tests replace normal references with black images, Gaussian noise, or randomly mismatched images. These conditions reduce or eliminate target-reference correlation and therefore probe whether the learned matcher and gate suppress unusable side information.

As shown in Fig. 7, degraded references increase bitrate and reduce PSNR relative to normal references. This agrees with the directional expectation that a reference with less shared structure is less useful, but it is not a numerical verification of Proposition 1. The component ablation in Table VI and the same-reference controls in Table VII provide the module-level evidence: reference acquisition, CLM, and CLS make distinct empirical contributions. We therefore use the theory only to describe the correlation trend, not to predict the gain of an individual module.

VII. CONCLUSION

We presented GRCL, a learned image coding framework that combines three routes for acquiring semantically correlated references with latent-domain matching, selective synthesis, and conditional entropy coding. The experiments show that stronger external references and the proposed CLM/CLS architecture make distinct contributions: acquisition enlarges the available side-information space, while matching and synthesis determine whether imperfect references improve rate-distortion performance. The theoretical result is a scoped analysis of correlated latent recovery; it supports the expected direction with reference correlation but is not an end-to-end guarantee for retrieval, generation, or entropy coding. The local-dictionary branch provides the lowest-latency deployment option, whereas web and diffusion branches trade service dependence and latency for access to a broader reference space. Future work includes faster or embedding-conditioned reference generation and broader private-set stress tests beyond the reported PHRD evaluation.

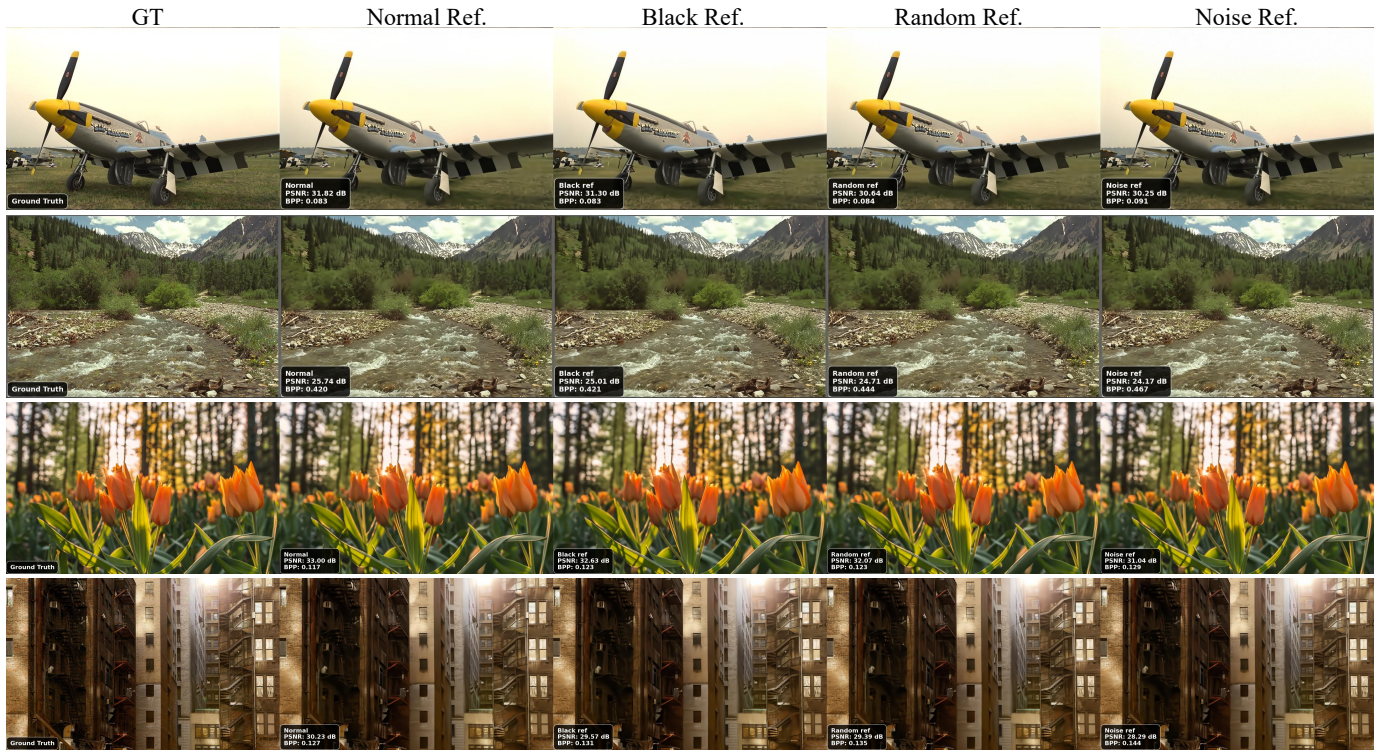
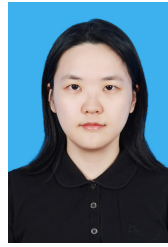


Fig. 7: Stress tests under four reference conditions: (a) normal references, (b) black references, (c) randomly mismatched references, and (d) Gaussian-noise references. Bitrate and PSNR are shown for each condition. These experiments test whether the implemented codec degrades gracefully; they provide qualitative evidence for the correlation trend but are not a direct numerical verification of the subspace bound.

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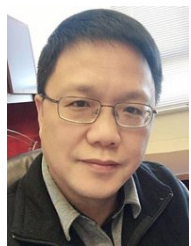


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